

ANUPREET SINGH

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EDUCATION

University of Maryland, Baltimore County

(Master of Science in Computer Science, GPA: 3.8/4.0)

Baltimore, MD

Aug 2024 – May 2026

Indian Institute of Information Technology, Sonapat

(Bachelor of Technology in Computer Science and Engineering, GPA: 8.45/10)

Haryana, India

Aug 2019 – May 2023

SKILLS / RELEVANT COURSEWORK

Programming Languages: Python, JavaScript/TypeScript, C/C++, Java, SQL

Tools and Frameworks: PyTorch, LangChain, CrewAI, Qdrant, Django, FastAPI, React, Next.js, MongoDB, PostgreSQL, PyTest, GoogleTest, AWS, Docker, Kubernetes, GitHub Actions(CI/CD), Claude Code

Areas of Interest and Coursework: AI & LLMs, Data Structures & Algorithms, Software Engineering (SDLC), Distributed System Design, Object Oriented Design (OOD), Computer Architecture

INDUSTRY EXPERIENCE

Data AI Intern | Platinum Business Services (R&D Team) | Catonsville, MD

Dec 2025 – Present

- Developed backend services for an AI-native GRC platform that automates compliance check of a system security plan draft with NIST 800-53, cutting reviews from **days of manual work to minutes**.
- Classified each control as fully, partially, or not satisfied by the uploaded plan and generated a gap report citing the evidence behind every verdict, replacing a manual control-by-control walkthrough.
- Built a parsing and chunking pipeline that splits **~1,200** controls into requirement, guidance, and assessment-objective sections, letting answers cite the exact sub-clause rather than the whole control.
- Implemented a fallback mechanism across 5 embedding models (OpenAI, Ollama) that re-runs RAG queries with insufficient control citation hits, lifting retrieval from **32/50 to 50/50** hand-reviewed compliance queries.
- Enforced a grounding contract that assembles citations only from retrieved chunks and short-circuits to an **Insufficient evidence** reply before any model call, so no compliance verdict rests on **fabricated control text**.
- Built the React interface for plan upload, gap-report review, and follow-up Q&A on unmet controls, along with exposing embedding and chat model selection options to users.
- Diagnosed an OOM bug where freshly pulled 4–5 GB LLM weights lingered in the OS page cache; fixed it with targeted `posix_fadvise` eviction, **eliminating model-load failures**.

Software Engineer (AI) Intern | PMA | Tallahassee, FL

Sep 2025 – Dec 2025

- Built a React + Django AI assignment-review platform to flag rubric mistakes, cutting coach prep from **5 to 2 hours**.
- Architected a fully serverless AWS stack using an SQS-triggered Lambda AI worker, DynamoDB, and S3/CloudFront, sustaining 200+ submissions across 5 cohorts at **~\$3/month** idle cost.

Full Stack Developer | Turfco Landscape Co. | Surrey, BC

Jan 2023 – June 2024

- Engineered an admin-configurable cost estimation engine, cutting quote turnaround from **~2 days** to **<1 min**.
- Shipped an admin pricing console that cut price-change lead time from a **developer deploy** to an **instant edit**, letting non-technical staff run seasonal updates unassisted.
- Implemented idempotent Stripe webhooks, moving deposit collection online with **zero** duplicate charges.
- Developed and shipped the React SPA with client-side routing, role-aware protected routes and form state validated against the same schema as the API, serving customer and admin experiences from one bundle.
- Sustained **300** req/s at **<100 ms** p95 latency on one EC2 instance via Redis caching and PostgreSQL indexing.

ACADEMIC EXPERIENCE

Graduate Assistant | UMBC - Computer Architecture & Software Engineering

Aug 2025 – June 2026

- Designed a C++ assignment on virtual memory (testing knowledge of MMU address translation, multi-level page tables, and kernel page-fault handling) and graded coursework in computer architecture for **40+** students.
- Mentored 10 student **Scrum** teams through sprint planning, reviews, and backlog refinement for their projects.
- Coordinated between 40 students and **stakeholders**, keeping all 10 projects on schedule amid evolving requirements.

Teaching Assistant | UMBC - Data Structures

Jan 2025 – June 2025

- Developed 20+ unit tests for the hash table project, automating logic checking and reducing grading time by **80%**.
- Helped **50+** students design and debug C++ projects implementing core data structures and graded homework.

PROJECTS

Bloodborne TAG | *Text Adventure Game with natural language command parsing*

- Engineered a **command parser** using few-shot Mistral-7B for slot extraction and embedding-similarity-based intent matching, mapping free-form playtest commands to 14 deterministic actions at **96%** intent accuracy.
- Architected a data-driven game engine that loads a **41-room** world and **60+** items/NPCs from JSON into in-memory session state, gating progression on generic key-item fields so new content ships with **zero** engine code changes.

When Pixels Talk Back | *Watermarking EMR Data to analyse impact on CV models*

- Fine-tuned **3 computer-vision models** (ResNet50, DenseNet169, MobileNetV2) for classification over COVID-19 lung CT scans via two-stage ImageNet transfer learning, to establish a baseline peaking at **93%** accuracy and **0.951** F1.
- Embedded a **1024-bit blind fragile watermark** into the Region-of-Non-Interest of CT scans via key-scrambled LSB embedding of a SHA-512 fingerprint, enabling content authentication with near-lossless quality at **SSIM 0.999**.
- Benchmarked **3 watermarking algorithms** across all 3 models, showing the fragile scheme costs **0%** accuracy loss, versus 11.2% for robust embedding and ~1% for reversible embedding with a 300-bit payload.

Nothing Lost in Translation | *Lexically constrained Sequence Generation*

- Built a **Grid Beam Search decoder** that guarantees positive/negative lexical constraints in any autoregressive sequence generation, demonstrated on English-to-Russian NMT with a **15-point BLEU** gain (25.45 $\rightarrow$ 40.59).
- Reconstructed a beam search decoder from scratch, outperforming HuggingFace's stock `.generate()` by **1.45 BLEU**.